

Data Ledger: Expert Specialization in Generation, Not Prefill

Companion to *Generation-Time Routing Reveals Expert Specialization That Prefill Measurements Miss* (Shorthill, v1.1, concept DOI 10.5281/zenodo.20779604). Compiled 2026-08-28 from the archived runs in `data/`. Every value below was re-read from the source tables on that date.

Significance labels. **XHIGH** the claim fails without it **HIGH** removes a main objection **MED** supports or scopes the claim **LOW** integrity check or worked example

1. Introduction

A mixture-of-experts (MoE) layer routes each token to a few of many expert blocks. The standard question is whether those experts divide up knowledge by subject. The best known answer, from the Mixtral report, is no. That answer was measured over supplied text, which the model processes in a single parallel pass (prefill). The model also routes tokens while it writes its answer, one token at a time (generation). Nobody had asked whether the two passes agree.

They do not. Over prefill, one expert wins nearly every subject. Over generation, each subject gets its own leading expert. The shift holds when prefill and generation are matched in length, and it repeats in a second, larger model whose expert numbering has nothing to do with the first. This ledger lists what was run, where each number lives, how much weight each number carries, and why the result matters.

2. What the experiments were

Three runs, all greedy decoding, all on 8-bit community fine-tunes of Qwen3.5 MoE models (HauhouCS "Uncensored Aggressive" variants), all captured with a patched llama.cpp binary that dumps router logits per token per layer. Routing weights are rebuilt the same way everywhere: softmax over 256 router logits, keep the top 8, renormalize to sum to one. For an expert, W is its mean weight over all tokens, S the fraction of tokens that select it, Q its mean weight when selected. A domain's winner is the expert with the highest W over that domain's prompts and all layers.

Run 1. 35B primary probe

60 short questions, 20 academic domains, 3 per domain. Qwen3.5-35B-A3B (40 layers, 256 experts, top-8, about 3B active). One capture per prompt, split at the prompt boundary: prompt tokens are the prefill block, produced tokens are the generation block. Generation capped at 2,056 tokens. The generation block was also analyzed twice, once trimmed at the first chat-template spill and once untrimmed.

Run 2. 35B token-balanced control

Same model. Three prompts, each packing twenty domain questions and padded to exactly 446 prompt tokens, each allowed 2,048 generated tokens. Purpose: check that the prefill/generation difference is not just a short-set versus long-set artifact. Compared the top-expert sets of the two passes (Jaccard overlap) and the per-token routing entropy of each pass.

Run 3. 122B replication

Same 60 prompts on Qwen3.5-122B-A10B (48 layers, 36 gated-DeltaNet and 12 softmax attention, 256 experts, top-8, about 10B active). Its expert indices are unrelated to the 35B model's. Two captures: a prefill-only pass (-n 0) and a generation pass (-n 2048). The generation pass also recorded its own prefill block.

3. Provenance

Early work; the 35B primary run predates the provenance discipline used later, so it has no binary hash. Paths are inside the paper repository's data/ folder. Git: github.com/jeffreywilliamportfolio/paper-expert-specialization-generation. Claim index: SOURCES.md.

Run 1

```
run_id      20260408T235839Z_domain_specialist_no_think_hauhau
model       Qwen3.5-35B-A3B-Uncensored-HauhauCS-Aggressive-Q8_0.gguf
decode      no-think, greedy, -n 2056
prompts     domain_specialist_probe_60_no_think.tsv (60 = 20 x 3)
cells       60/60; prompt tokens 2404, generated 110412, total 112816
capture bin capture_activations (custom llama.cpp example); hash NOT recorded
files       35b-60prompt-primary/results_domain_specialists_20260408T235839Z.{md,json}
            35b-60prompt-primary/raw/...npz (per-domain x layer x expert W/S/Q, 20x40x256;
            NOT raw per-token tensors)
```

Run 2

```
run_id      20260410T173400Z_domain_expert_probe_3chunk_balanced_gen_n2048
model       Qwen3.5-35B-A3B-Uncensored-HauhauCS-Aggressive-Q8_0.gguf
decode      generation+prefill, seed 42, temp 0, top-k 1, ctx 32768, flash-attn, KV q8_0, -n
            2048
prompts     3 chunks (20A, 20B, 20C), 446 prompt tokens each; padding 0 / 73 / 44 tokens
capture bin /workspace/consciousness-experiment/capture_activations; hash NOT recorded
files       35b-3chunk-token-balanced/DOCS/RESULTS.md (cited values)
            35b-3chunk-token-balanced/results/*.json|.md, per_token_*/ (per-token npz+tsv)
            35b-3chunk-token-balanced/METHOD/ (analysis scripts, capture_activations.cpp,
            qwen_router.py)
```

Run 3

```
prefill run 20260412T160341Z_qwen122_domain_specialist_probe_60_routing_only (-n 0)
gen run     20260412T161833Z_qwen122_domain_specialist_probe_60_gen_n2048 (-n 2048)
model       Qwen3.5-122B-A10B-Uncensored-HauhauCS-Aggressive-Q8_K_P.gguf (144,514,683,488
bytes; sha256 not recorded)
capture bin sha256 c3e205b3a6324f6ce11e775b8ab1c9115904e348ec118fcb62d5e31ccae62e70 (both
runs)
flags       -ngl 999 -c 16384 -t 16 -fa on --cache-type-k q8_0 --cache-type-v q8_0 --seed 42
--temp 0 --top-k 1 --routing-only
files       122b-replication/routing-only/results.md + RESULTS/*.json (prefill-only)
            122b-replication/generation/results.md, text-results.md + RESULTS/*.json
(generation, with own prefill block)
            122b-replication/JOURNAL-122B.md
```

4. The data, with significance

Run 1: 35B primary

Datum	Where	Label	Note
Prefill: expert 224 wins 18 of 20 domains; 3 distinct winners (103 linguistics, 130 chemistry); Herfindahl 0.815, normalized entropy 0.13	primary .md, Prefill > Domain winners	XHIGH	the prefill half of the claim
Generation: 20 distinct winners across 20 domains; Herfindahl 0.050, entropy 1.00	primary .md, Generation Trimmed > Domain winners	XHIGH	the generation half
Generalist 224 wins exactly one domain in generation (political science, W 0.0092)	primary .md line 1802	HIGH	the generalist does not merely share; it drops out
Trimmed and untrimmed generation give the same 20 winners	RESULTS-summary line 53; .md lines 924-948 vs 1782-1806	HIGH	rules out a chat-template spill artifact
224 is also the top expert overall on prefill (W 0.0127, S 0.079)	primary .md, Prefill > Overall	MED	the generalist is a genuine default, not a tie-break winner
Philosophy: generation winner E114 (W 0.0188, S 0.097); E114 was rank 5 within philosophy over prefill	primary .md lines 84, 942	MED	worked example of an expert rising into a domain
All 60 cells spill into chat-template continuation (60 layer-39 trim events)	RESULTS-summary line 28	LOW	limitation; handled by the trim check above
W = S x Q residual at most 5.55e-17 on every token set	RESULTS-summary line 34	LOW	bookkeeping integrity
Capture totals: 60/60 cells, 2,404 prompt tokens, 110,412 generated	RESULTS-summary lines 21-26	LOW	completeness

Run 2: 35B token-balanced control

Datum	Where	Label	Note
Jaccard overlap of prefill vs generation top-expert sets, by W: 0.000 / 0.231 / 0.067 (chunks A/B/C); by S: 0.067 / 0.231 / 0.000	DOCS/RESULTS.md, Jaccard table	XHIGH	the shift survives matched token count
Mean per-token routing entropy (top-8, normalized by log2 8): prefill 0.9575, generation 0.9527; per-chunk deltas all negative (-0.0046, -0.0062, -0.0037)	DOCS/RESULTS.md, Routing Entropy; METHOD/qwen_router.py line 9	HIGH	kills the "generation just routes more diffusely" reading
Prompt tokens exactly 446 per chunk; padding 0 / 73 / 44	results/...token_audit_summary	MED	the control is actually balanced
E114 W-rank prefill to generation: A 30 to 21, B 82 to 25, C 89 to 7	DOCS/RESULTS.md, Expert 114	LOW	not used in the paper; separate line of work

Run 3: 122B replication

Datum	Where	Label	Note
Generation (trimmed): 18 distinct winners across 20 domains; E159 and E40 each win two; Herfindahl 0.060, entropy 0.95	generation/results.md lines 1804-1827	XHIGH	replication of the generation half
Prefill (routing-only run): E233 wins 13 of 20; 7 distinct winners; Herfindahl 0.445, entropy 0.42. This is the value in the paper.	routing-only/results.md, Domain Winners	HIGH	replication of the prefill half; see flag
Prefill (the generation run's own prefill block): E233 wins 15 of 20; 6 distinct; Herfindahl 0.575, entropy 0.32. Not in the paper.	generation/results.md lines 88-112	HIGH	paired reading; see flag
Global top experts change identity between passes: prefill E233, E45, E72, E9, E215; generation E0, E11, E5, E1	routing-only Overall; generation Overall top	MED	shift is not only at the domain-winner layer
Expert indices unrelated across models: 40 vs 48 layers, different layouts; generalist is 224 in 35B and 233 in 122B	JOURNAL-122B.md line 5; routing-only line 7	MED	makes the replication a shape replication, not an index match
49 of 60 cells hit the 2,048 cap; mean generated length 1,868; every cell spills into chat-template continuation	generation/text-results.md lines 18-22	MED	limitation; stated in the paper
Mean prompt tokens 39.37	routing-only/results.md line 37	LOW	"a prompt is tens of tokens"
Philosophy generation winner E40 (W 0.0099): outside the top 12 by W in philosophy prefill (W 0.0062), but rank 1 on the archive's composite specialist score for philosophy prefill	generation line 1822; routing-only lines 286-290; RESULTS json domains_summary.philosophy	LOW	worked example; see flag
E114 (the 35B philosophy winner) in 122B: computer science rank 5 (prefill) / 4 (generation); nowhere else above rank 17	routing-only line 125; generation line 1812	LOW	indices do not carry across models

External corroboration (found 2026-08-28)

Datum	Where	Label	Note
Wang, Hayou and Nalisnick, "The Myth of Expert Specialization in MoEs" (JHU, arXiv 2604.09780, April 2026), five other MoE models: "prefill-phase expert usage is not a reliable proxy for full generation usage pattern"; "during prefilling, semantically unrelated sequences can activate exactly the same experts"; "characterizing specialization requires observing complete trajectories." They do not test whether generation routing sorts by domain.	arxiv.org/abs/2604.09780, Sections 5.2, 5.3, Fig. 7	HIGH	independent support for the premise, on non-Qwen and non-hybrid models; not yet cited in v1.1

5. Flags found in this audit

122B prefill is unpaired and not bit-stable. The paper's 122B prefill row (13/20, 7 distinct) comes from the routing-only run. The generation run captured its own prefill with the same binary hash, prompts, and flags, differing only in `-n`, and there E233 wins 15/20 with 6 distinct. The claim holds under either reading (both concentrated; generation 18 distinct regardless). But the paper's sentence that prefill routing "is deterministic given the prompt and the weights" is contradicted by this archive, and Table 1 pairs a prefill from one capture with a generation from another without saying so. Fix: report the paired 15/20 or keep 13/20 and disclose the separate capture and the gap.

E40 example cuts both ways. "Mid-pack in philosophy prefill" is true by W and false by the archive's composite score (rank 1). Either give the W rank or drop the example.

Smaller wording issues. "Each attending to everything before it" is wrong for the 30/40 and 36/48 gated-DeltaNet layers. Hayashi et al. use the official FP8 checkpoint, not "the same checkpoint we use." Per-token entropy (0.958) is never defined as top-8 normalized. The 35B `.npz` holds per-layer summaries, not raw router tensors.

6. Why it matters

The negative result was measured in the wrong pass. The most-cited evidence that MoE experts do not specialize by topic was taken over supplied text, which is prefill routing. Prefill routing in these models is dominated by one default expert regardless of subject. Read the same prompts over generation and each subject gets its own leader. So a prefill-only or pooled measurement understates specialization by construction, and a negative prefill result says nothing about generation.

It is a regime effect, not a length effect. With 446 prompt tokens against 2,048 generated tokens per chunk, the two passes still pick nearly disjoint expert sets, and per-token entropy barely moves. Generation spreads each token over about as many experts as prefill does and sends it to different ones. The leader identity changes; the spread does not. That is the specialization signal.

It replicates where it should. The 122B model has different depth, a hybrid DeltaNet/attention stack, and unrelated expert numbering. Its generalist is a different expert and its per-domain leaders are different experts, yet the concentrated-then-dispersed shape is the same. A second group, working on five unrelated models, independently found that prefill usage does not predict generation usage. The premise is not a quirk of one quantized fine-tune.

Practical consequence. Any expert-naming pipeline, specialization benchmark, or ablation study that scores routing over supplied text inherits the prefill picture. Measure the two passes separately and read specialization from generation. The XHIGH rows above are enough to carry that recommendation; the HIGH rows close the obvious objections; the flags are fixable in a revision without touching any XHIGH value.